

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following cycles has the highest thermal efficiency for the same compression ratio? A) Otto cycle B) Diesel cycle C) Dual cycle D) Rankine cycle
- 2 The primary reason CI engines are more efficient than SI engines is due to: A) Higher speed B) Higher compression ratio C) Use of carburetor D) Better fuel quality
- 3 Delayed ignition timing in SI engines leads to: A) Reduced power output B) Increased knocking C) Higher thermal efficiency D) Early peak pressure
- 4 Which of the following factors does not significantly affect brake thermal efficiency? A) Compression ratio B) Ambient humidity C) Fuel-air ratio D) Ignition timing
- 5 MPFI stands for: A) Multi Pressure Fuel Injection B) Multiple Pressure Fuel Injection C) Maximum Power Fuel Injection D) Multi Point Fuel Injection
- 6 In diesel engines, fuel is injected: A) At the beginning of compression B) Just after TDC of compression stroke C) At the end of exhaust stroke D) During the power stroke
- 7 Abnormal combustion leads to: A) Engine wear and power loss B) Increased thermal efficiency C) Better emission characteristics D) Low peak pressure
- 8 The brake thermal efficiency of an engine is the ratio of: A) Indicated power to fuel energy B) Brake power to input fuel energy C) Mechanical efficiency to indicated power D) Brake power to heat rejection
- 9 SI engines are generally preferred for: A) Heavy trucks B) High-speed passenger cars C) Tractors D) locomotives
- 10 The function of an EGR system is to: A) Reduce unburned HCB) Cool intake air C) Increase oxygen supply D) Lower combustion temperature to reduce NOx
- 11 Which fuel property significantly reduces knocking in SI engines? A) High cetane number B) Low boiling point C) High octane number D) Low viscosity
- 12 The use of oxygenated fuels such as ethanol-blended petrol leads to reduced CO emissions primarily because: A) Oxygenated fuels have lower energy content B) They increase ignition delay C) They promote more complete oxidation of carbon during combustion D) They form a richer air-fuel mixture

- Q2. Solve the following.
- A) Explain the working of a four-stroke SI engine with a neat valve timing diagram. 6
 - B) Differentiate between SI and CI engines with respect to construction and working principles. 6
- Q3. Solve the following.
- A) Describe in detail the phenomenon of ignition delay in a compression ignition (CI) engine. What are the key factors that influence the duration of the ignition delay period? 6
 - B) Describe the operation of wet sump lubrication system with the help of neat sketch 6
- Q4. Solve Any Two of the following.
- A) Describe the working of a simple carburetor and its limitations in modern SI engines. 6
 - B) What is supercharging in IC engines? Discuss the advantages and limitations of supercharging with respect to engine performance and efficiency. 6
 - C) Explain Battery Ignition System with neat diagram. 6
- Q5. Solve Any Two of the following.
- A) Explain the stages of combustion in SI engines and the factors affecting flame propagation. 6
 - B) Explain the phenomenon of knocking in spark ignition (SI) engines. What are the major factors that contribute to knocking, and what techniques can be employed to minimize or control it? 6
 - C) What are the various methods used to control emissions in internal combustion engines? Explain the role of Exhaust Gas Recirculation (EGR) in emission reduction with neat sketch. <https://www.batuonline.com> 6
- Q6. Solve Any Two of the following.
- A) Explain the need for alternative fuels and list their advantages over conventional fuels. 6
 - B) What are the major pollutants emitted by IC engines and what are their health and environmental impacts? 6
 - C) Draw and explain layout of electric vehicles. 6

*** End ***